



Bharatiya Vidya Bhavan's

SARDAR PATEL INSTITUTE OF TECHONOLOGY

(An Autonomous Institute Affiliated to University of Mumbai)
MUNSHI NAGAR, ANDHERI (WEST), MUMBAI - 400058

MID SEMESTER EXAMINATION

Sr.No. 8513

AY:	2025-26	Session:	Odd MSE October 2025	Date:	08-10-2025
Sem:	1	Slot:	10:00am - 11:00am	C. Code:	CS413
Exam Type:	Written	Block:	401	Seat No:	2022600040
Course Section:	1				

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Name of the candidate. Atharva Pramod Patil.

Candidate's UID number: 2 0 2 2 6 0 0 0 4 0

Examination class room number: 4 0 1

Invigilator's signature with date: [Signature] 8/10/25

(To be filled in by the candidate)

EXAMINATION: BY Btech. MSF

PROGRAM: Deep learning, AI/ML

DAY AND DATE: 8/10/25 Wed

COURSE NAME: Deep learning

SEMESTER : 7

I / II / III / IV / V / VI / VII / VIII

(To be filled in by the examiner)

Question numbers	1	2	3	4	5	6	Total	Signature of the examiner
Marks awarded								
Marks after revaluation								

INSTRUCTIONS TO CANDIDATES

- 1) Candidate should ensure that all answer books including supplementary answers books received by them bear the signature of the junior supervisor, otherwise the answer books will not be examined.
- 2) Begin the answer to each question on a new page. For each answer, write the corresponding question number and sub question number if any in the respective column provided in the answer sheet.
- 3) Do not write anything in the column provided for the marks to be assigned by the examiners.
- 4) Candidate will not be permitted to leave the examination hall until an hour after the question papers are distributed.
- 5) Every candidate present must sign against their seat number on the attendance sheet provided by the junior supervisor.
- 6) Candidates are forbidden to (i) bring any book, notes. Scribbling papers, mobile telephones, programmable calculators, or any other similar devices.(ii) speak or communicate in any manner to any other candidate, while the examination is in progress, and (iii) take with them any answer-book written or blank while leaving the examination hall. The supervisors/authorized persons are authorized to check the students.
- 7) Candidate suspected to be guilty of any of the aforesaid acts will be allowed to write their paper only after giving an undertaking in writing that the decision of the College Examination Authorities in respect of the reported act of unfair means is binding on them.
- 8) Candidates should write their answers legibly. They are warned that **zero marks** will be assigned to **answers which cannot be assessed by the examiners owing to illegible handwriting**.
- 9) Write on both side of a page. Rough work, when necessary, should be done on the Left- hand side and in pencil only.
- 10) While underlining of answers for focusing attention is permitted, use of varied inks, except for illustration and figures must be avoided.
- 11) No sheet shall be torn from the answer-books provided nor shall additional papers attached to them.
- 12) The answer-books will be scrutinized before they are sent to examiners.
- 13) All answer-books supplied shall be returned whether written or blank.
- 14) Nothing shall be written on the question-paper.
- 15) If candidate needs anything, they should approach the Junior Supervisor without disturbing other candidates. However, they should not leave their seat on my account.
- 16) A warning bell will be given ten minutes before the close of the examination. Candidates will not be allowed to leave the examination hall during the last ten minutes. At the final bell, they must stop writing and be ready to hand over their answer-books to the Junior Supervisor. They should not leave their seats until answers-books from all candidates are collected by the Junior Supervisor.
- 17) A candidate who disobeys any instructions issued by the Senior/Junior Supervisor or who is guilty of rude or disobedient behavior is liable for disciplinary action to be taken against him/her by the College Examination Authorities.

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Q.1

a)

$$f(w) = \frac{1}{2} \|xw - y\|^2.$$

gradient update rule is as follows

$$w_i \leftarrow w_i + \Delta w_i$$

$$\text{Here } \Delta w_i = -\eta \frac{\partial L}{\partial w}.$$

Here. ~~$\frac{\partial L}{\partial w}$ can be given~~
 L can be given as.

~~$$L = \frac{1}{2} \sum (\hat{y}_i - y_i)^2$$~~

~~$$\therefore \frac{\partial L}{\partial w} = \frac{1}{2} \sum \frac{\partial (\hat{y}_i - y_i)^2}{\partial w}$$~~

~~$$\frac{\partial L}{\partial w} = \frac{2}{2} (\hat{y} - y)$$~~

$$L = \frac{1}{2} \sum (y_e - y_o)^2.$$

$$\frac{\partial L}{\partial w} = \frac{1}{2} \frac{\partial (y_e - y_o)^2}{\partial w}$$

$$\frac{\partial L}{\partial w} = \frac{2}{2} (y_e - y_o) \frac{\partial (y_e - y_o)}{\partial w}$$

$$\frac{\partial L}{\partial w} = (y_e - y_o) \frac{\partial y - \partial xw}{\partial w \cdot \partial w}$$

$$\therefore \frac{\partial L}{\partial w} = (y_e - y_o) \cdot x_i$$

$$\frac{\partial L}{\partial w} = -x_i (y_e - y_o).$$

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Now for convergence the $\frac{dL}{dx} = 0$.

$$\therefore 0 = -n(y_e - y_0)(x).$$

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	Q.1	
	b)	i) Overfitting occurs when the model learns to much out of the given dataset. ii) the This usually means that the model is not capable enough to understand the variance out of the dataset. iii). This usually happens when the bias of the dataset is large along with the variance. iv) As a result the model performs overfitting over the given dataset. v) This could be tackled by performing regularization. vi) Regularization is a concept where you can penalize the entire training dataset for errors over high bias. vii) Example: L1 regularization, L2 regularization as penalties over loss function. the These overall penalties are hence a function of variance hence removing overfitting.

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Q.2a) For an input size of $H \times W \times C$, the number of multiplication goes as follows.

$\therefore$ The output layer size is a function of this

$$H_0 = \left\lfloor \frac{H}{F} \right\rfloor + 1$$

$$W_0 = \left\lfloor \frac{W}{F} \right\rfloor + 1$$

These happens for k independent kernels over the same period.

$\therefore$ The total number of multiplications come out to be.

$$H_0 \times W_0 \times F \times \frac{C}{F}$$

Here $\frac{C}{F}$ is the number of ^{depth} ~~kernels~~.

it is going to take one at a time.

for depth wise separable convolution

each depth layer independly goes through the kernel. Hence the number of multiplications come out to be

$$H_0 \times W_0 \times F \times C$$

Hence the depthwise separable provides more insight compared to the normal convolution.

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Q.2

b). i) Undercomplete autoencoders are autoencoders with a small ~~masked~~ sized masked layer.

ii) This ~~leads to~~ forces the model to learn critical information out of the dataset.

iii) However it could also lead to the missing out on important information.

i) Overcomplete autoencoders are autoencoders with a larger sized masked layer.

ii) As a result, a lot of times the model ends up picking up the input layer ~~as~~, exactly same as to that of the output layer.

iii) This ~~can lead to a~~

As a result the learning in output and the input is ~~as for~~ an identity function.

This is because, the outcome is inclusive of the input set. This leads ~~an~~ the output to be a function as

$$\text{Output} = f(\text{input}) + \text{some change.}$$

Hence the learning is

$$\text{Output} - \text{input} = f(\text{input}) - \text{input} + \text{some change}$$

Here $f(\text{input}) - \text{input}$ tends to be an identical function.

Coming back to under complete autoencoders, the dimension of ~~input~~ ^{output} is smaller than that of input.

$$\text{output} = f(\text{input} - b)$$

∴ Learning rate makes a change in dimension resulting it to equal a PCA.

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Q.30) $\frac{\partial L}{\partial h_t} = ?$

The forget gate in LSTM is as follows

$$f_t = \sigma(w_f(h_{t-1}, x) + b_f)$$

∴ $\frac{\partial L}{\partial f_t}$

$$f_{t-k} - f_t = \sigma_1 w_f h_{t-k} + \sigma_1 w_f x_{t-k-1} + \sigma b_f - \sigma w_f h - \sigma w_f x_t - \sigma b_f$$

$$f_{t-k} - f_t = \sigma_1 w_f (h_{t-k} - h) + \sigma w_f (x_{t-k} - x_t)$$

$$\therefore L = \frac{1}{2} (f_{t-k} - f_t)^2$$

$$= \frac{1}{2} (\sigma_1 w_f ((h_{t-k} - h) + x_{t-k} - x_t))$$

$$\therefore \frac{\partial L}{\partial h_{t-k}} = \frac{\partial L}{\partial h_{t-k}}$$

for this case assume $k=1$

$$\therefore \frac{\partial L}{\partial h_{t-1}} = \frac{1}{2} \sigma_1 w_f (1 - h + x_{t-1} - x_t)$$

Here you realise the gradient descent does not lead to vanishing rate it strictly defines the loss 0

$$= \frac{1}{2} (\sigma_1 w_f (h \cdot f_t (h_{t-1}) \dots - h))$$

$$\therefore \frac{\partial L}{\partial h_{t-k}} = \frac{1}{2} \sigma_1 w_f (h \cdot f_t (h_{t-1}) \dots - h)$$

This is a constant term.
As a result LSTM nullifies his recursion to prevent vanishing gradient.

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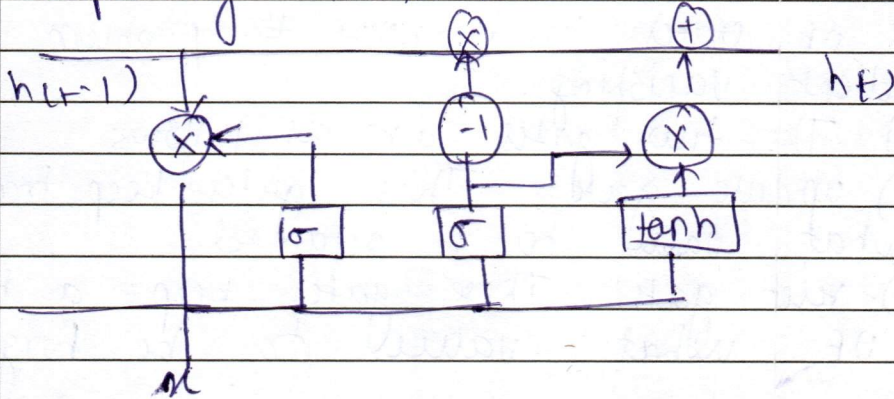
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	Q.3	
	b).	i) The GRU is a variation of RNN that makes use of gates to overcome the problem of gradient vanishing. ii) The two gates are as follows a) update gate: These gates keep track of what value to be retained b) reset gate: These gates keep a track of what values to be forgotten
		Update gate: Let the weight for x for update be w_u, with bias b_u. We use sigmoid function to limit the values to either 0 or 1. $u_t = \sigma(w_u(h_{t-1}, x) + b_u)$
		Similarly for reset gate: Let the weight for x for reset be w_r with bias b_r. $r_t = \sigma(w_r(h_{t-1}, x) + b_r)$
		Now the candidate function is now given as $\tilde{c} = \tanh(w_c \cdot (r_t \times h_{t-1}, x) + b_c)$
		Here $\tanh$ restricts the values from -1 to 1. A reset is multiplied to h_{t-1}, signifying if a memory block of h_{t-1} has to be forgotten.
		∴ Now the final, $c = u_t \times \tilde{c} + (1 - u_t)(h_{t-1})$
		Here the c can either be an update for the

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reset properly or reset for an old one depending on the case.



Hence the number of hyperparameters are

 w_r w_u w_c b_r b_u b_c

Making a total of 6 parameters.

LSTM on the other hand makes use of three gates.

forget gate, input gate and output gate.

Moreover there is a candidate function.

Hence the number of parameters here turn out to be 8.

as the eqns are as follows:

$$f_x = \sigma(w_f(h_{t-1} \parallel x) + b_f)$$

$$i_x = \sigma(w_i(h_{t-1} \parallel x) + b_i)$$

$$o_x = \sigma(w_o(h_{t-1} \parallel x) + b_o)$$

$$\tilde{c} = \tanh(w_c(h_{t-1} \parallel x) + b_c)$$

Hence this proves that GRU is significantly faster than LSTM.

Hence applications which make use of large number of parameters, something like

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